IT5603 - Distributed and Cloud Computing

Assignment 2
Web Services using Python

Submitted by

Name	**Yuvaraj K**
Register Number	**2022506079**

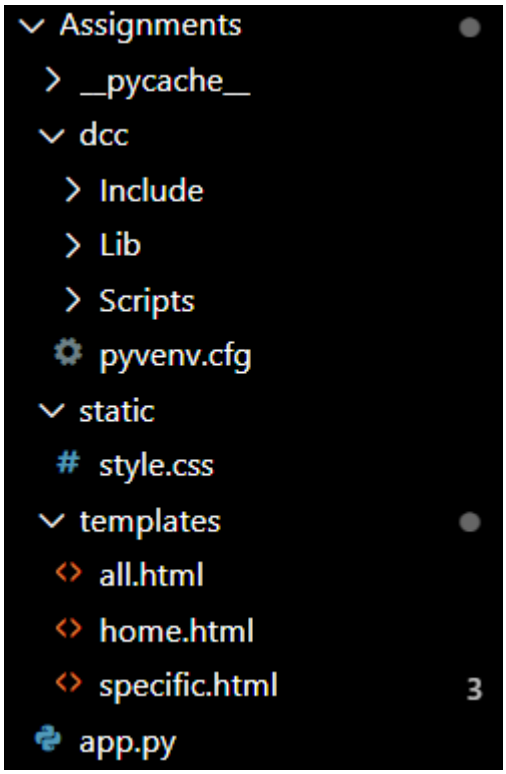
Department of Information Technology
Madras Institute of Technology,
Anna University, Chennai – 600 044

January 2025 – May 2025

Web Services using Python

Application: Temperature Checker
Provides temperatures of a specific city, or all cities.

Project Structure



home.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Temperature Checker</title>
   <link rel="stylesheet" href="/static/style.css">
</head>
<body>
   <header>
      <div class="header-container">
         <h1>Temperature</h1>
         <div class="link-group">
            <a href="specific">Specific City</a>
            <a href="all">All Cities</a>
         </div>
      </div>
   </header>
   <p>
      Welcome to the Temperature Checker application!
      <br><br>
      This is your go-to tool for checking the temperatures of cities across the
country.
      <br><br>
      With a user-friendly interface, this application allows you to do two
different things.
      You can either select a city from a list and instantly view its current
temperature.
      Or, you can view the temperatures of all available cities at once.
      So, go ahead! Explore!

   </p>
</body>
</html>
```

specific.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Temperature Checker</title>
   <link rel="stylesheet" href="/static/style.css">
</head>
<body>
   <header>
      <div class="header-container">
         <h1>Temperature</h1>
         <div class="link-group">
            <a href="specific">Specific City</a>
            <a href="all">All Cities</a>
         </div>
      </div>
   </header>
   <main class="container">
      <form action="/specific" method="POST" class="form">
         <select name="city" required>
            <option value="" disable select>Select City</option>
            <option value="Chennai">Chennai</option>
            <option value="Pune">Pune</option>
            <option value="Bengaluru">Bengaluru</option>
            <option value="Mumbai">Mumbai</option>
            <option value="New Delhi">New Delhi</option>
            <option value="Hyderabad">Hyderabad</option>
            <option value="Calcutta">Calcutta</option>
         </select>
         <button type="submit" class="submit-btn">Find</button>
      </form>
   </main>
   <p style="{{ style }}">{{ message }}</p>
</body>
</html>
```

all.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Temperature Checker</title>
    <link rel="stylesheet" href="/static/style.css">
</head>
<body>
    <header>
        <div class="header-container">
            <h1>Temperature</h1>
            <div class="link-group">
                <a href="specific">Specific City</a>
                <a href="all">All Cities</a>
            </div>
        </div>
    </header>
    <p>{{ message|safe }}</p>
</body>
</html>
```

style.css

```
body {
    font-family: Arial, sans-serif;
    margin: 0;
    padding: 0;
    background-color: #f5f5f5;
}

.header-container {
    background-color: #2283d3;
    color: white;
    padding: 1rem;
    display: flex;
    justify-content: space-between;
    align-items: center;
}

.header-container h1 {
    margin: 0;
    font-size: 1.5rem;
}

.link-group {
    display: flex;
    gap: 1rem;
    height: 100%;
}

a {
    color: white;
    text-decoration: none;
    font-weight: 500;
    display: flex;
}

a:hover {
    color: rgb(216, 207, 207);
    height: 100%;
}

.container {
    display: flex;
    justify-content: center;
```

```
    align-items: center;
    margin: 2rem auto;
    padding: 2rem;
    background-color: #ccebf1;
    border-radius: 10px;
    max-width: 600px;
    box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
}

.form {
    display: flex;
    flex-direction: column;
    width: 100%;
}

.form select {
    padding: 0.5rem;
    margin-top: 0.3rem;
    border: 1px solid #ccc;
    border-radius: 5px;
    font-size: 1rem;
    background-color: #fff;
    color: #333;
    cursor: pointer;
    transition: border-color 0.3s, box-shadow 0.3s;
}

.form select:focus {
    outline: none;
    border-color: #2b92e6;
    box-shadow: 0 0 5px #2b92e6;
}

.form select::-ms-expand {
    display: none;
}

.form select:disabled {
    background-color: #e9e9e9;
    cursor: not-allowed;
    color: #888;
}

.submit-btn {
```

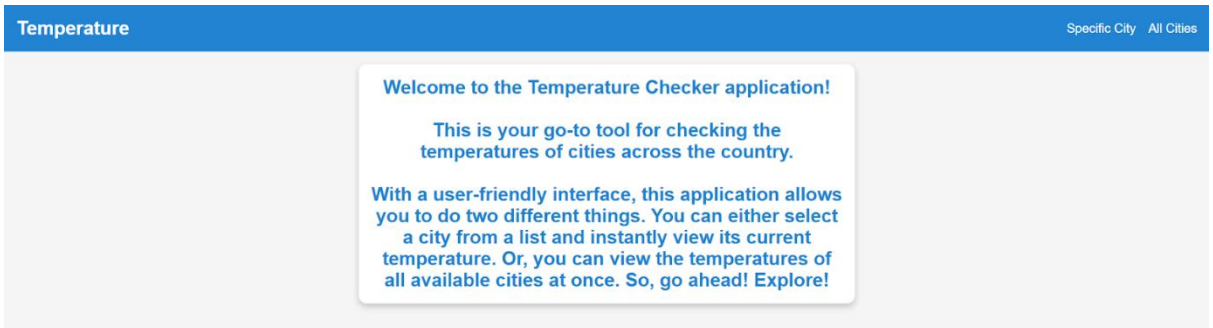
```
    margin-top: 1.5rem;
    background-color: #2283d3;
    color: white;
    border: none;
    padding: 0.8rem 1.2rem;
    font-size: 1rem;
    cursor: pointer;
    border-radius: 5px;
}

.submit-btn:hover {
    background-color: #2b92e6;
}

p {
    font-size: 1.5rem;
    font-weight: bold;
    color: #2283d3;
    text-align: center;
    padding: 1rem;
    margin: 1rem auto;
    background-color: #ffffff;
    border-radius: 10px;
    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
    max-width: 600px;
}
```

app.py

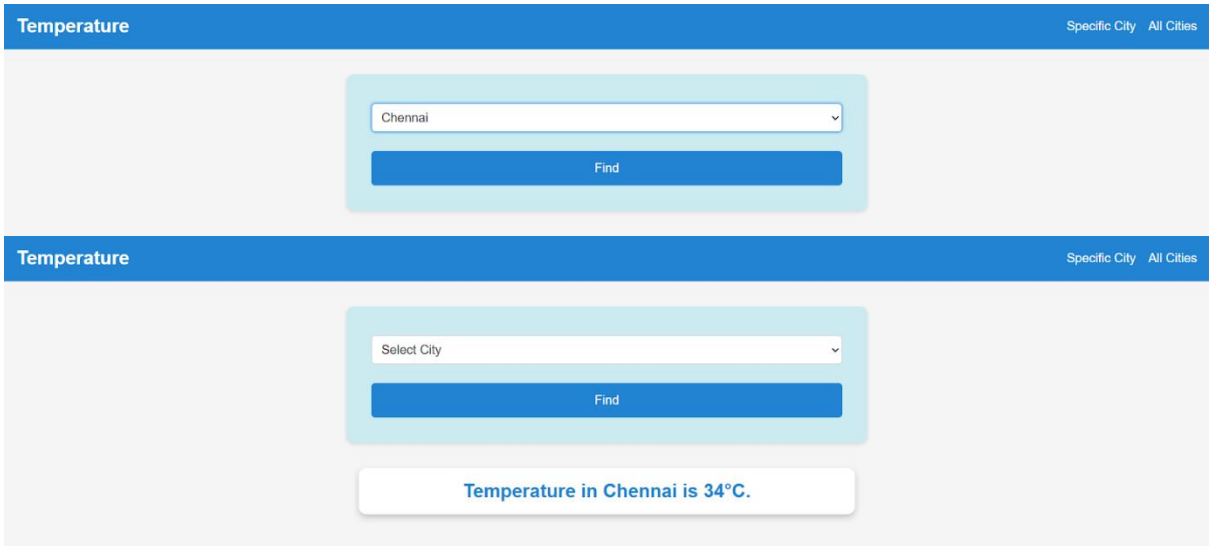
```
from flask import Flask, render_template, request

temperatures = {
    "Chennai": 34,
    "Pune": 29,
    "Bengaluru": 27,
    "Mumbai": 32,
    "New Delhi": 36,
    "Hyderabad": 33,
    "Calcutta": 31
}
app = Flask (__name__)

@app.route('/')
def home():
    return render_template('home.html')

@app.route('/specific', methods=['GET', 'POST'])
def specific():
    m = ""
    s = "display: none;"
    if request.method == "POST":
        city = request.values.get("city")
        s = "display: block;"
        m = f"Temperature in {city} is {temperatures[city]}°C."
    return render_template('specific.html', message=m, style=s)

@app.route('/all')
def all():
    m = "Temperatures in All Cities<br><br>"
    for city in temperatures:
        m += f"{city}: {temperatures[city]}°C<br>"
    return render_template('all.html', message=m)

if __name__ == '__main__':
    app.run(debug=True)
```

Output

Home Page



Specific City



All Cities